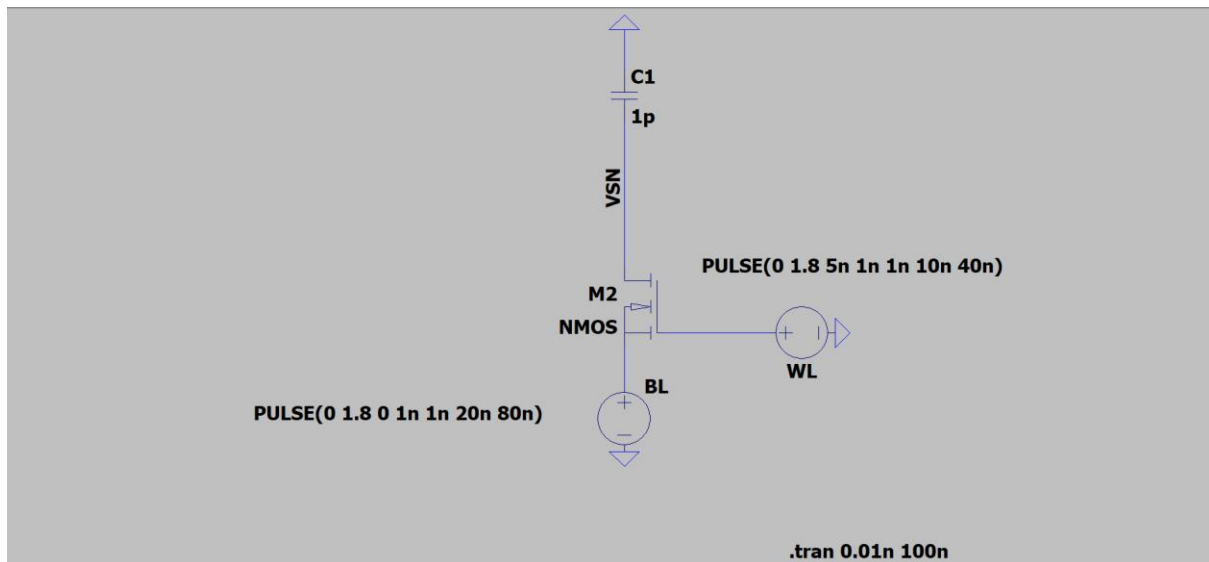
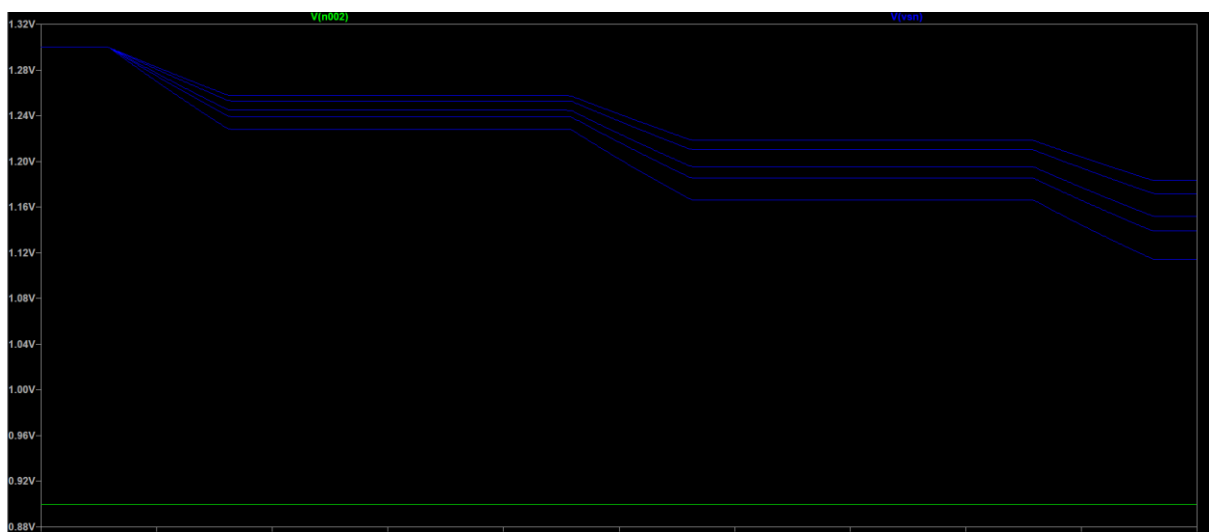
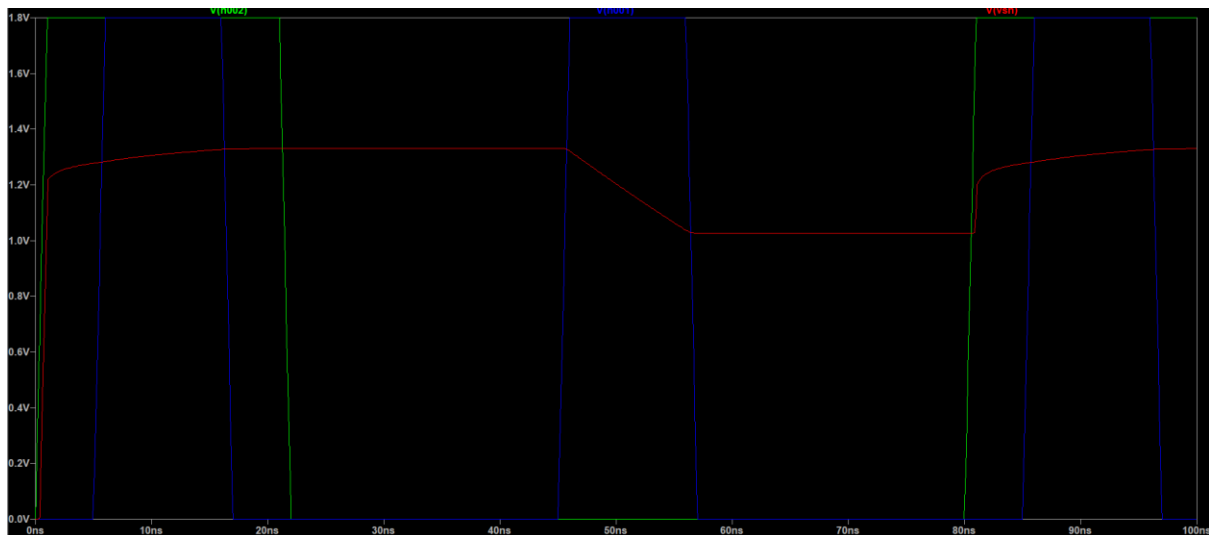
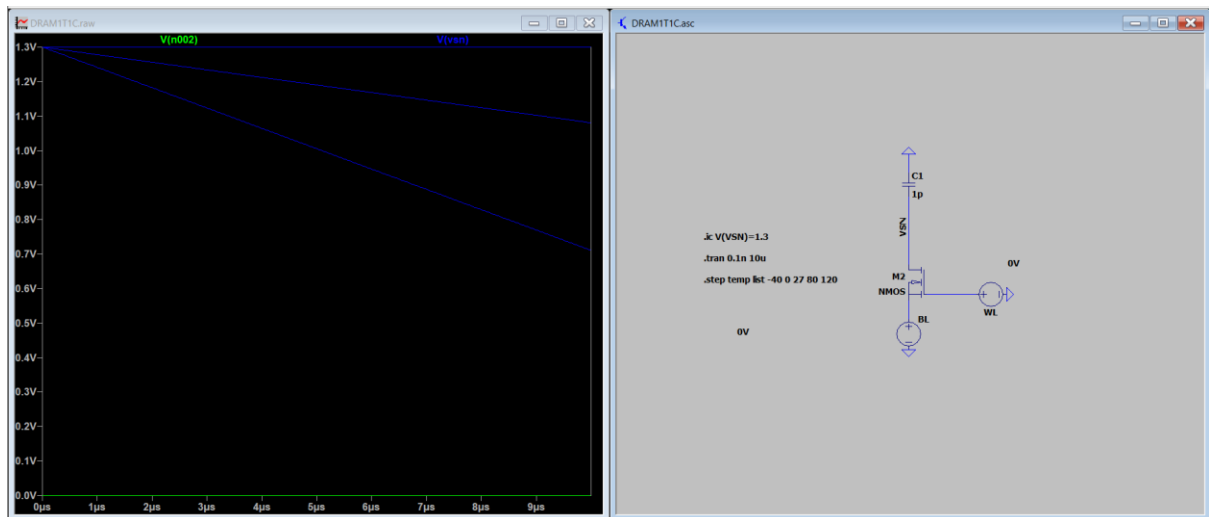
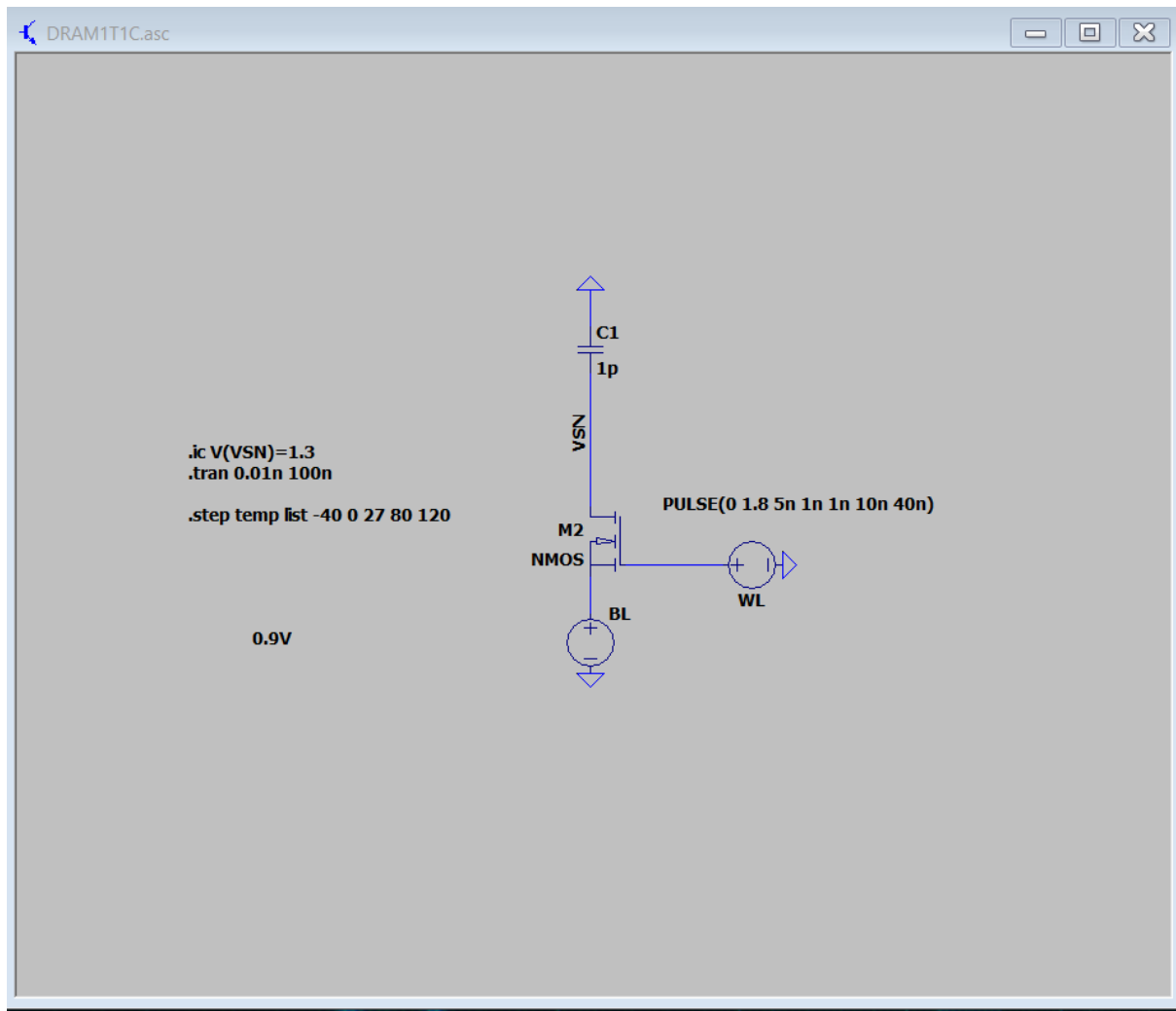




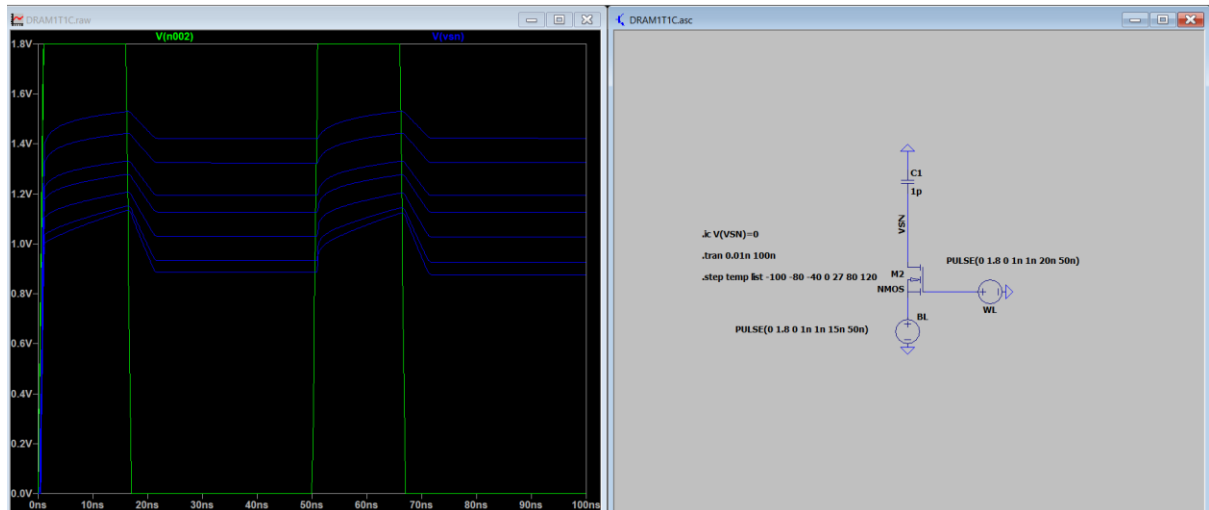
Basic DRAM 1T 1C Circuit



For read in each temp(-40 0 27 80 120)



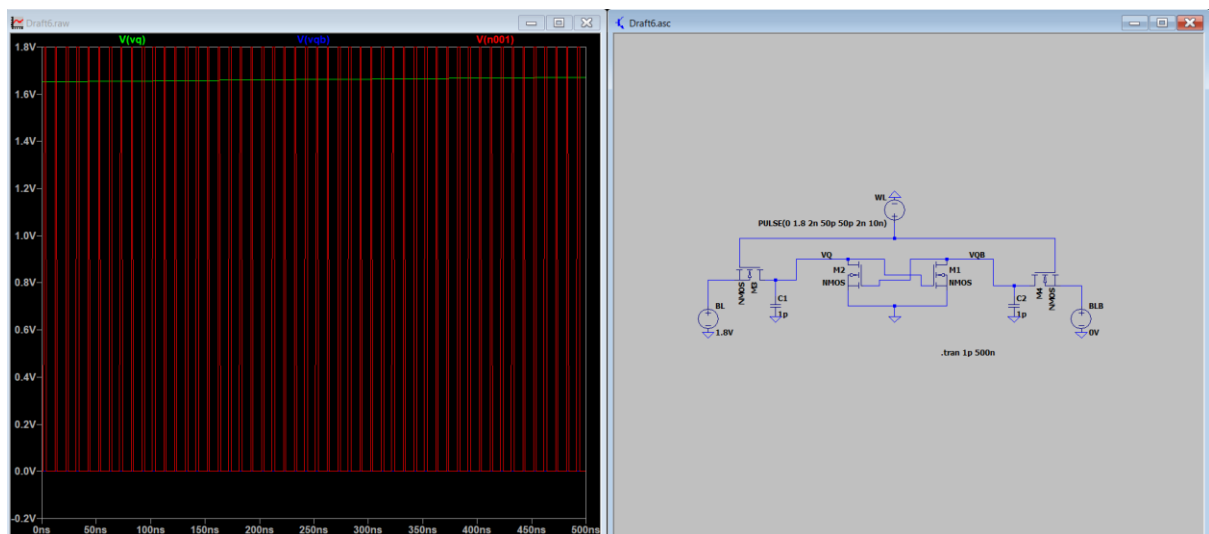
Charge retention VSN vs temp



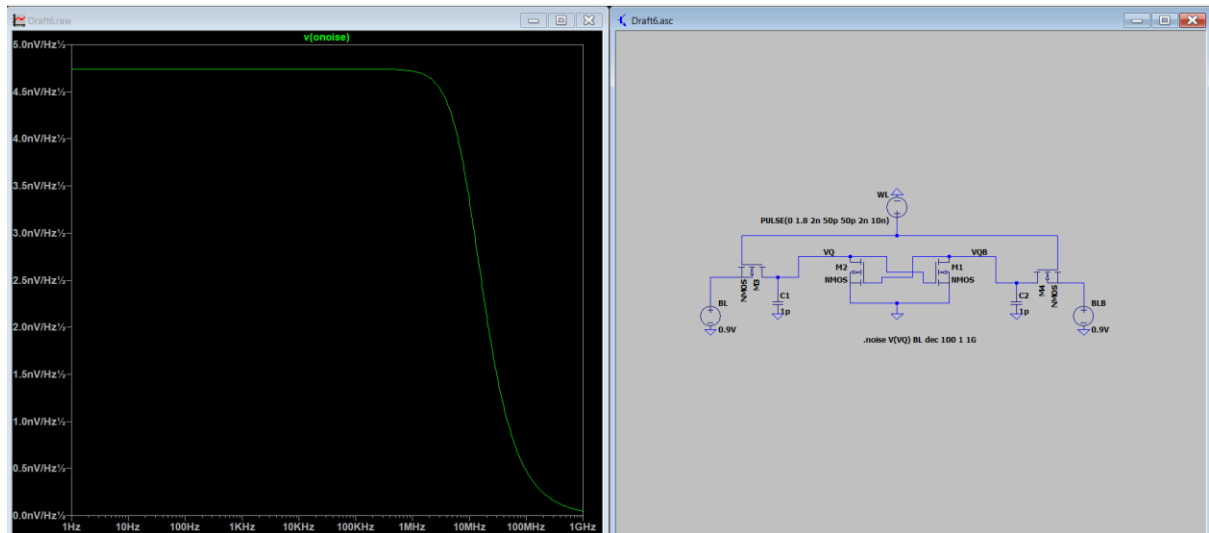
For Write margin VBL and VSN for corresponding temps



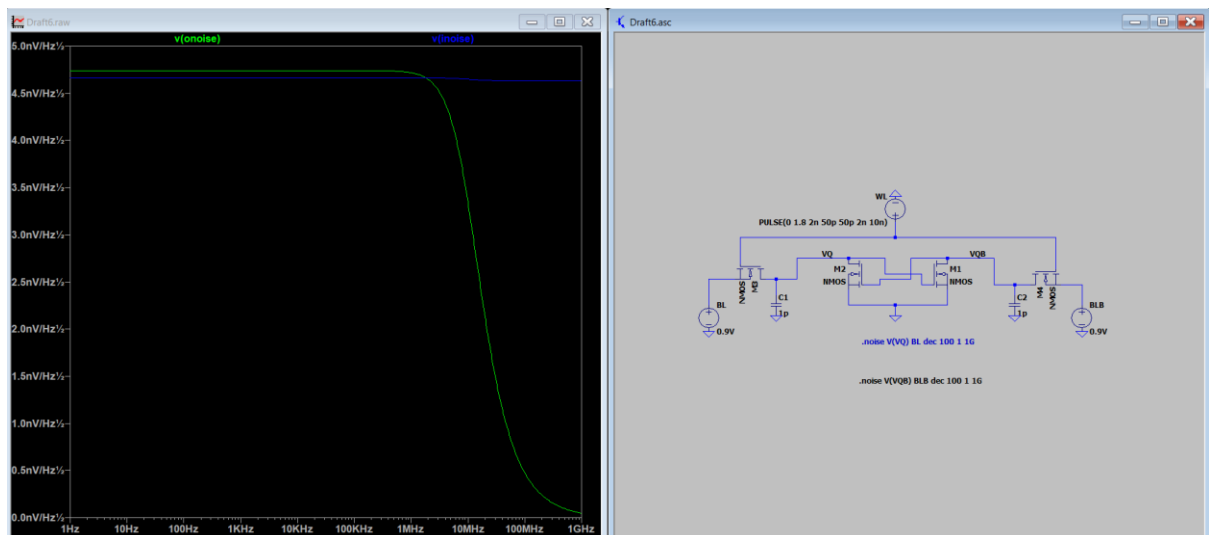
Power analysis



For 4t DRAM read



Noise analysis of 4t DRAM



SNR

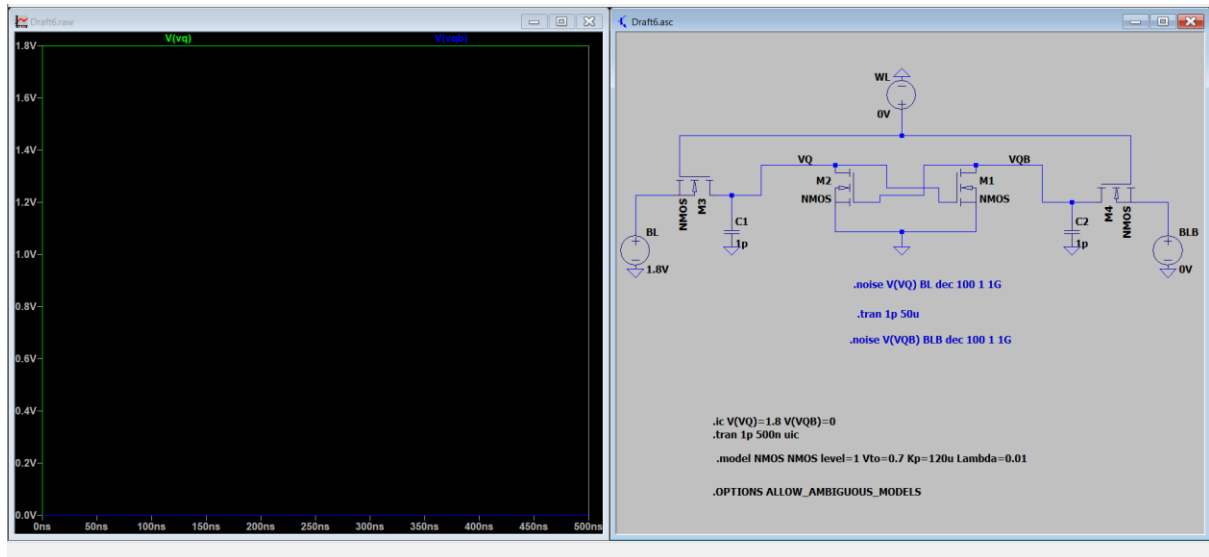
Flat noise(onoise)= 4.7

Corner freq= almost 1MHz

Bandwidth= 1MHz

Integrated output noise= 4.7

$$\text{SNR} = 20 \cdot \log_{10}(1.65 / (4.7 / 10^6)) = 110.9 \text{ dB}$$



Retention test : ability to hold a value(ideal case)

Also for 12nm